FOR A COHEN-MACAULAY MORPHISM THE SHRIEK OF THE STRUCTURE SHEAF IS A FLAT MODULE IN ONE DEGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BV8 (sources/stacks-duality/pinned-original-source.tex line 5978).

Allowed standard prerequisites (editorial).

- The relative dimension of a Cohen-Macaulay morphism is locally constant. Morphisms, Lemma labelled morphisms-lemma-flat-finite-presentation-CM-fibres-relative-dimension, cited at proof lines 6015-6017. Fixes d.
- Coherence of the shriek of the structure sheaf and flatness of its single cohomology sheaf. Same chapter, Lemmas labelled lemma-shriek-coherent and lemma-flat-shriek, cited at proof lines 6017-6025. Give coherence and flatness once vanishing is known.
- The affine line computation and the quasi-finite case. Same chapter, Lemmas labelled lemma-shriek-affine-line and lemma-flat-quasi-finite-shriek, cited at proof lines 6033-6039. Settle the two model cases of the reduction.
- Characterisation of Cohen-Macaulay morphisms through factorization. More on Morphisms, Lemma labelled more-morphisms-lemma-flat-finite-presentation-characterize-CM, cited at proof lines 6029-6033. Produces the factorization with g quasi-finite flat.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 045R: sources/stacks-chow/tags.txt, source lines 5446–5446. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02NM: sources/stacks-chow/tags.txt, source lines 3485–3485. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5993 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AU1: sources/stacks-duality/pinned-original-source.tex, source lines 4686–4686. Same-chapter pointer into the author environment lemma at source line 4686; invoked at proof line 5995 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BV6: sources/stacks-duality/pinned-original-source.tex, source lines 5915–5915. Same-chapter pointer into the author environment lemma at source line 5915; invoked at proof line 5996 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BV7: sources/stacks-duality/pinned-original-source.tex, source lines 5962–5962. Same-chapter pointer into the author environment lemma at source line 5962; invoked at proof line 6003 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BUU: sources/stacks-chow/tags.txt, source lines 14887–14887. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6009 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AA1: sources/stacks-duality/pinned-original-source.tex, source lines 4609–4609. Same-chapter pointer into the author environment lemma at source line 4609; invoked at proof line 6010 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 0F42 let $f : X \to Y$ be a morphism of FTS_S. If f is Cohen-Macaulay (More on Morphisms, Definition 045R), then*

$$f^!\mathcal{O}_Y = \omega_{X/Y}[d]$$

for some coherent $\mathcal{O}_X$-module $\omega_{X/Y}$ flat over Y where d is the locally constant function on X which gives the relative dimension of X over Y.

Proof. The relative dimension d is well defined and locally constant by Morphisms, Lemma 02NM. The cohomology sheaves of $f^!\mathcal{O}_Y$ are coherent by Lemma 0AU1. We will get flatness of $\omega_{X/Y}$ from Lemma 0BV6 if we can show the other cohomology sheaves of $f^!\mathcal{O}_Y$ are zero.

The question is local on X, hence we may assume X and Y are affine and the morphism has relative dimension d. If $d = 0$, then the result follows directly from Lemma 0BV7. If $d > 0$, then we may assume there is a factorization

$$X \xrightarrow{g} \mathbf{A}^d_Y \xrightarrow{p} Y$$

with g quasi-finite and flat, see More on Morphisms, Lemma 0BUU. Then $f^! = g^! \circ p^!$. By Lemma 0AA1 we see that $p^!\mathcal{O}_Y \cong \mathcal{O}_{\mathbf{A}^d_Y}[-d]$. We conclude by the case $d = 0$. □